Homework 2

Homework 2 involves simple analysis of a fasta sequence file, the minimal lingua franca for raw DNA sequence. The goals of this exercise are to gain familiarity with reading files line by line, using loops, and making logical tests.

FastA files are used to exchange nucleotide and protein sequence data. The FastA file consists of two kinds of line.

1. doc lines - exactly one line per sequence, beginning with the > character.
    - Whatever follows the >, up to the first space, is the sequence ID
    - The remainder of the line, up to the newline, is free-text documentation. The documentation is optional
2. sequence lines – all other lines are sequence information for the sequence identified in the preceding docline. There may be many lines of sequence information.

Instructions

1. Pull from the *master* branch to your personal branch. This will create a folder called HW2.
2. HW2 includes both these instructions as well as a partially written template you can use as your starting point (hw2_template.py). Before starting, copy the template to a new name, preferably hw2.py (this makes it easy for me to identify your completed code).
3. After pulling, double check you are on your personal branch. If you are not use *Git->branches* to checkout your local branch
4. Write a program to do the following
    - Open and read the sequence file line-by-line as shown in class
    - Count and report the number of lines in the file
    - Count and report the number of sequences in the file (lines beginning with >)
    - Count and report the total number of sequence characters in all sequences. Do not count newline characters
    - Your result should exactly match the sample result in the file, hw2_output.txt
5. If you don't use *Push and Commit*, as suggested in the instructions, you will need to push your final result to your personal branch on the course repository. Do not push to *master*.

Hints

- Use the code I showed in class to open and read the file, not methods you may see that use *with.*
- Don't forget that every line read from the file contains the invisible newline (\n) character. These can be removed with the rstrip() function (see string functions).